

CATHERINE CHENG

cc2862@cornell.edu | (480) - 562 - 7503 | [PORTFOLIO](#) | [LINKEDIN](#)



EDUCATION

Cornell University, College of Engineering, Ithaca, NY
Bachelor of Science, Electrical and Computer Engineering

B.S. Expected May 2028

Relevant Coursework: Computer Architecture, Computer Systems Programming, Digital Logic, Digital System Design - Microcontrollers, Electronic Circuits, Embedded Systems, Microelectronics, Power Electronics, Signals and Systems

TECHNICAL SKILLS

- **Programming & HDLs:** C/C++, Python, Perl, SystemVerilog/Verilog, Bash/Linux, Makefile, CMake
- **Digital & Embedded:** RTL Design, FPGA, ARM Cortex-M, RP2350, Interrupts/ISRs, State Machines, UART, SPI, I2C, PWM
- **Hardware & Equipment:** PCB/Avionics Wiring, Soldering, Oscilloscope, DMM, Signal Generator, Power Supply
- **Software & Tools:** Intel Quartus Prime, ModelSim, Verilator, Git/GitHub, LTspice, PLECS, Betaflight, AutoCAD, Valor

PROJECTS

FPGA TinyRV1 Processor

August 2025 – December 2025

SystemVerilog, RTL Design, FPGA, Quartus, Verilator

Ithaca, NY

- Designed and implemented a subset of the RISC-V ISA in a single-cycle processor and specialized accumulate accelerator in SystemVerilog on a Terasic DE0-CV FPGA
- Built processor datapath and control logic, including register file, ALU, memory interfaces, and instruction decoding
- Verified processor functionality through directed, randomized, and X-propagation testing, exercising the full supported instruction set and validating synthesized hardware with Quartus and oscilloscope measurements.

Birdsong Synthesizer

August 2026 – September 2026

C, RP2350, SPI, DDS

Ithaca, NY

- Built a programmable birdsong synthesizer on an RP2350, allowing users to generate, record, replay, and sequence custom bird-like sounds using a keypad and slide potentiometer
- Implemented real-time audio generation using DDS at 50 kHz, transmitting synthesized samples over SPI to an external DAC and speaker
- Developed interrupt-driven recording and accelerated playback; measured 2.16 μ s ISR execution time against a 20 μ s sampling period to verify real-time performance

UART Number Guessing

April 2026 – May 2026

C, FRDM-KL46Z, UART, Interrupts/ISRs, Concurrency

Ithaca, NY

- Developed a two-board embedded game on FRDM-KL46Z microcontrollers using bidirectional UART and interrupt-driven communication to synchronize game state and user inputs
- Implemented synchronization and concurrency logic to coordinate shared resources and asynchronous events across the embedded system
- Integrated LCD displays, GPIO buttons, and PWM controlled LEDs with interrupt driven control to provide responsive user interaction and real-time game feedback

EXPERIENCE

Northrop Grumman

June 2026 – August 2026

Computer Engineer Intern

Baltimore, Maryland

- Developed a Perl-based tool to automate, verify, and standardize PWB design data processing, reducing a 25+ minute manual workflow to 4 minutes while limiting engineer involvement to final inspection
- Reevaluated outdated dielectric withstand voltage (DWV) requirements through hardware testing and analysis, supporting a safe reduction in test voltage to reduce unnecessary PWB stress and improve manufacturing yield
- Automated component-selection verification, reducing verification time 90% from 2 hours to 12 minutes and establishing the workflow as the Power group's standard for parts orders

Autonomous Drones Project Team

August 2025 – present

Electrical Subteam

Ithaca, NY

- Debugged power-distribution issues on IARC competition drones where components were not receiving sufficient current, redesigning electrical wiring to improve power delivery while reducing wiring complexity
- Designed system-level wiring schematics and integrated power, signal, and flight-control interfaces
- Performed electrical bring-up and debugging, tracing wiring and connection issues through continuity checks, measurements, and subsystem testing

Entertainment Engineering and Design

January 2025 – May 2026

Electrical Subteam

Ithaca, NY

- Built motorized Flappy Bird game using stepper motor and driver circuit to control obstacle movement timing
- Assembled and soldered motion control electronics for animatronic arm mechanism, integrating servo motors for coordinated movement
- Developed Arduino C++ PID velocity-control algorithms to achieve repeatable motor motion and coordinated movement sequences